

1.9 Connecting Multiple Representations of Limits

Calculus

Name: _____

CA #1

Evaluate each limit.

1. $\lim_{x \rightarrow 3^-} \frac{|x-3|}{x-3}$

2. $\lim_{x \rightarrow 4^+} \frac{|x-4|}{x-4}$

3. $\lim_{x \rightarrow -5^-} \frac{x+5}{|x+5|}$

4. $\lim_{x \rightarrow -6^+} \frac{|x+6|}{x+6}$

5. Let f be a function where $\lim_{x \rightarrow 2} f(x) = 4$. Which of the following could represent the function f ?

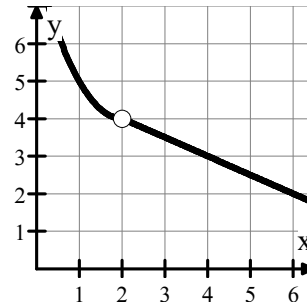
a.

$$f(x) = \begin{cases} \frac{x^2-4}{x-2}, & x \neq 2 \\ 3, & x = 2 \end{cases}$$

b. The two “pieces” are linear.

x	1.8	1.9	1.999	2	2.001	2.1	2.2
$f(x)$	7.2	7.1	7.001	4	3.999	3.9	3.8

c.



6. If g is a piecewise function with two linear “pieces” such that $\lim_{x \rightarrow 6} g(x)$ does not exist, which of the following could be representative of the function g ?

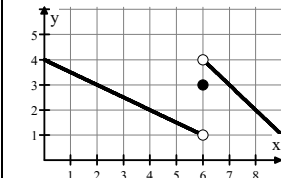
a.

$$g(x) = \begin{cases} 7-x, & x < 6 \\ 2x-14, & x > 6 \end{cases}$$

b.

x	3	4	5	6	7	8	9
$g(x)$	$\frac{3}{2}$	2	$\frac{5}{2}$	2	1	-1	-3

c.



1. -1	2. 1	3. -1	4. 1	5a. yes	5b. no	5c. yes	6a. Yes	6b. No	6c. Yes
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Answers to 1.9 CA #1